THE LOWER SHRIEK ALONG A LOCALIZATION COMMUTES WITH FINITE CONNECTED LIMITS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The equivalence between localized sheaves and slices. Same chapter, Lemma lemma-essential-image-j-shriek. Identifies the lower shriek with the projection from the slice, which is then compared.
- The description of the lower shriek on presheaves. Same chapter, Lemma lemma-describe-j-shriek. Gives the alternative direct argument, since sheafification is exact.

Source reference. Original lemma 00Y1: sources/stacks-sites/pinned-original-source.tex, source lines 5613–5613. Navigation pointer invoked at proof line 5613; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03CD: sources/stacks-sites/pinned-original-source.tex, source lines 5578–5578. Navigation pointer invoked at proof line 5578; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00XS: sources/stacks-sites/pinned-original-source.tex, source lines 5641–5641. Navigation pointer invoked at proof line 5641; the counted span contains the pointer only and the cited result is not reproduced inside it, so

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Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $U \in \mathrm{Ob}(\mathcal{C})$. The functor $j_{U!}$ commutes with fibre products and equalizers (and more generally finite connected limits). In particular, if $\mathcal{F} \subset \mathcal{F}'$ in $Sh(\mathcal{C}/U)$, then $j_{U!}\mathcal{F} \subset j_{U!}\mathcal{F}'$.*

Proof. Via Lemma 00Y1 and the fact that an equivalence of categories commutes with all limits, this reduces to the fact that the functor $Sh(\mathcal{C})/h_U^\# \to Sh(\mathcal{C})$ commutes with fibre products and equalizers. Alternatively, one can prove this directly using the description of $j_{U!}$ in Lemma 03CD using that sheafification is exact. (Also, in case $\mathcal{C}$ has fibre products and equalizers, the result follows from Lemma 00XS.) □